Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Building a Dashboard - Analyzing Sales, Revenue, Profit and Product Performance

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Experiment No.: 3
Building a Dashboard: Analyzing Sales, Revenue, Profit and Product Performance

GitHub Repository Link: *https://github.com/anuskaGHS/Business-Analytics*

Problem Statement

A retail company relies on manually prepared reports to monitor business performance. As a data analyst, develop a Power BI dashboard to visualize and monitor sales, revenue, profit, and product performance in an interactive and efficient manner.

Methodology / Steps Followed

Step 1: Data Collection

The AdventureWorks dataset was used for this experiment. The dataset consists of multiple files covering sales transactions, customer information, product details, territory data, and calendar information. The following data sources were identified:

- Sales Data (2020, 2021, 2022) - stored in MySQL database
- Customer Lookup - CSV flat file
- Product Lookup - CSV flat file
- Territory Lookup - CSV flat file
- Calendar Lookup - CSV flat file

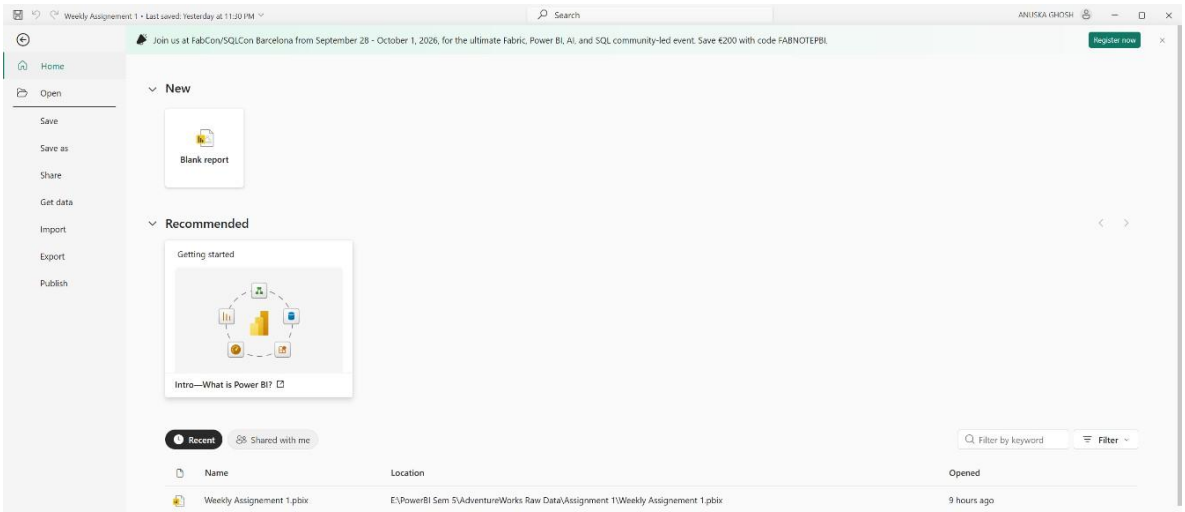


Figure 1: Power BI Desktop Home Screen

Step 2: MySQL Database Setup

A new database named AdventureWorks was created in MySQL Workbench 8.0. The Sales Data from three CSV files (2020, 2021, 2022) was imported into a single table named Sales using the Table Data Import Wizard. A total of 56,046 records were successfully loaded into the MySQL database, representing transactional sales data across three years.

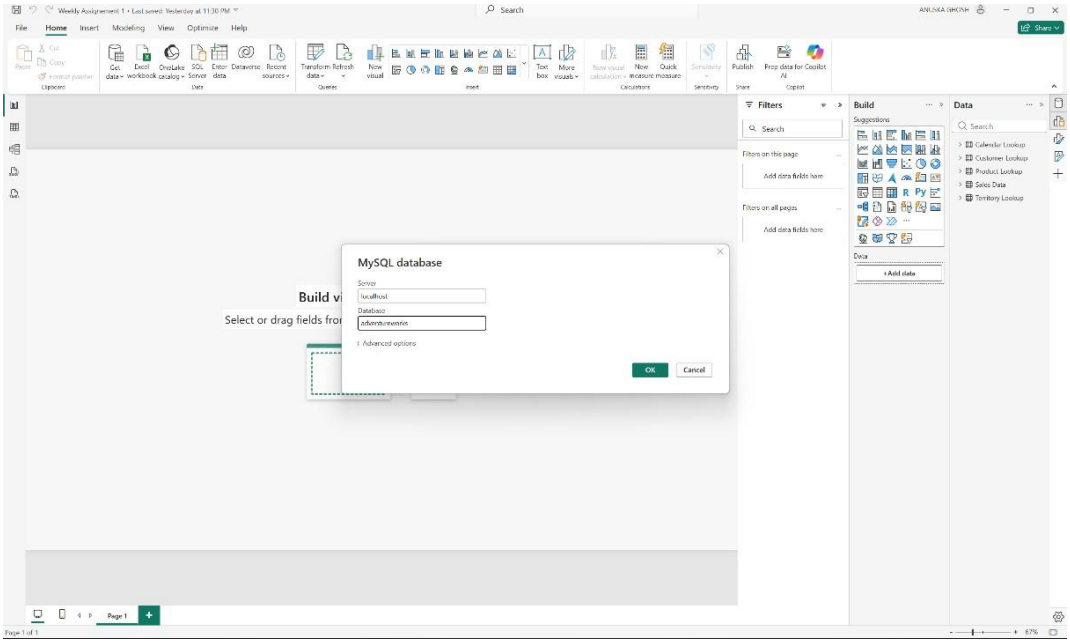


Figure 2: MySQL Database Connection in Power BI

Step 3: Data Import in Power BI

Power BI Desktop was opened and data was imported from two types of sources using the Get Data option:

- **MySQL Database** - Server: localhost, Database: adventureworks. The Sales table containing 56,046 records was loaded into Power BI.
- **CSV Flat Files** - Customer Lookup, Product Lookup, Territory Lookup, and Calendar Lookup files were imported directly using the Text/CSV connector.

The data types of OrderDate and StockDate columns in the Sales table were converted from Text to Date format in Power Query Editor.

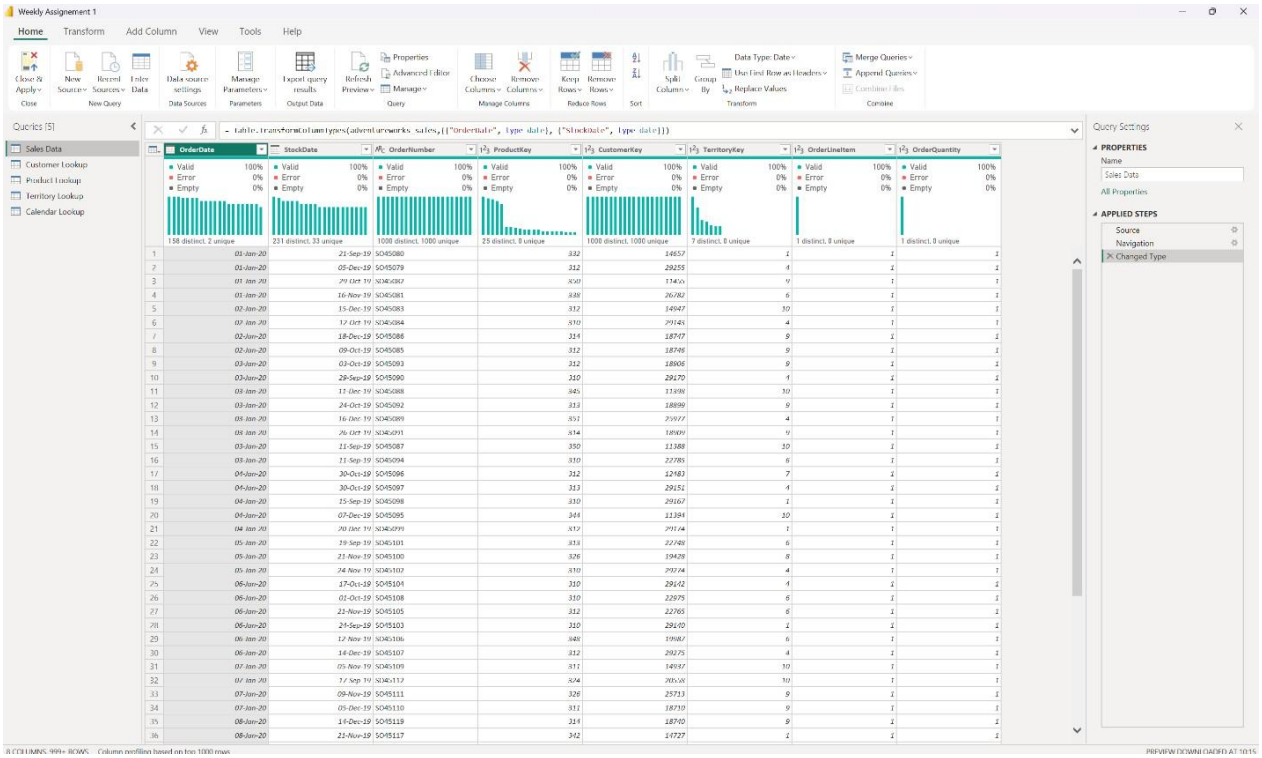


Figure 3: Sales Data Table

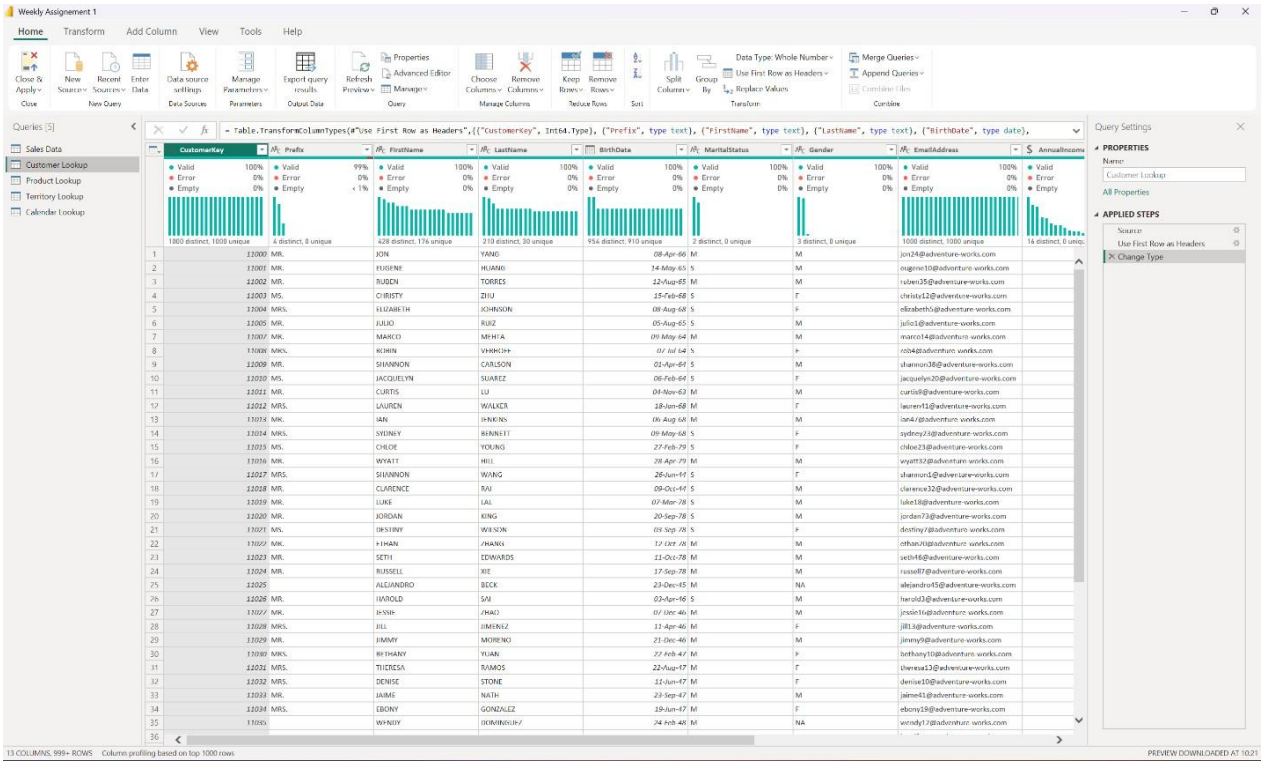


Figure 4: Customer Lookup Table

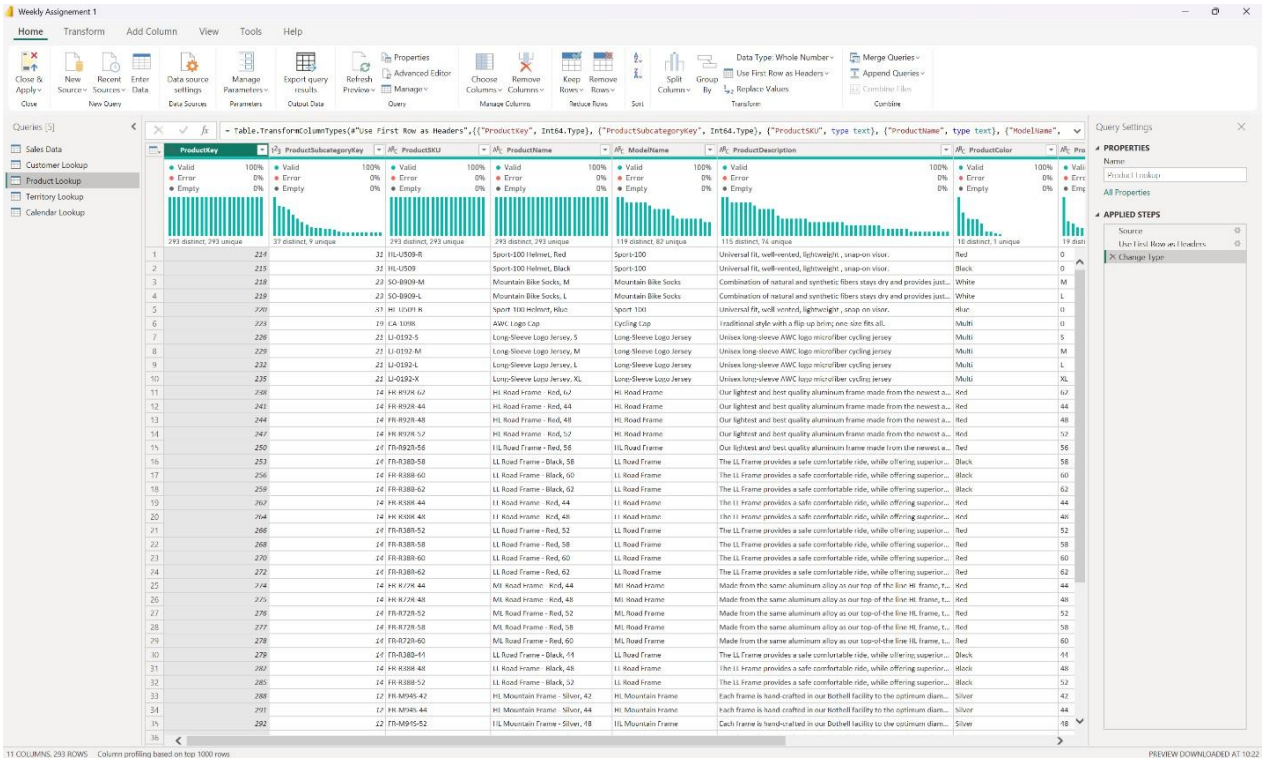


Figure 5: Product Lookup Table

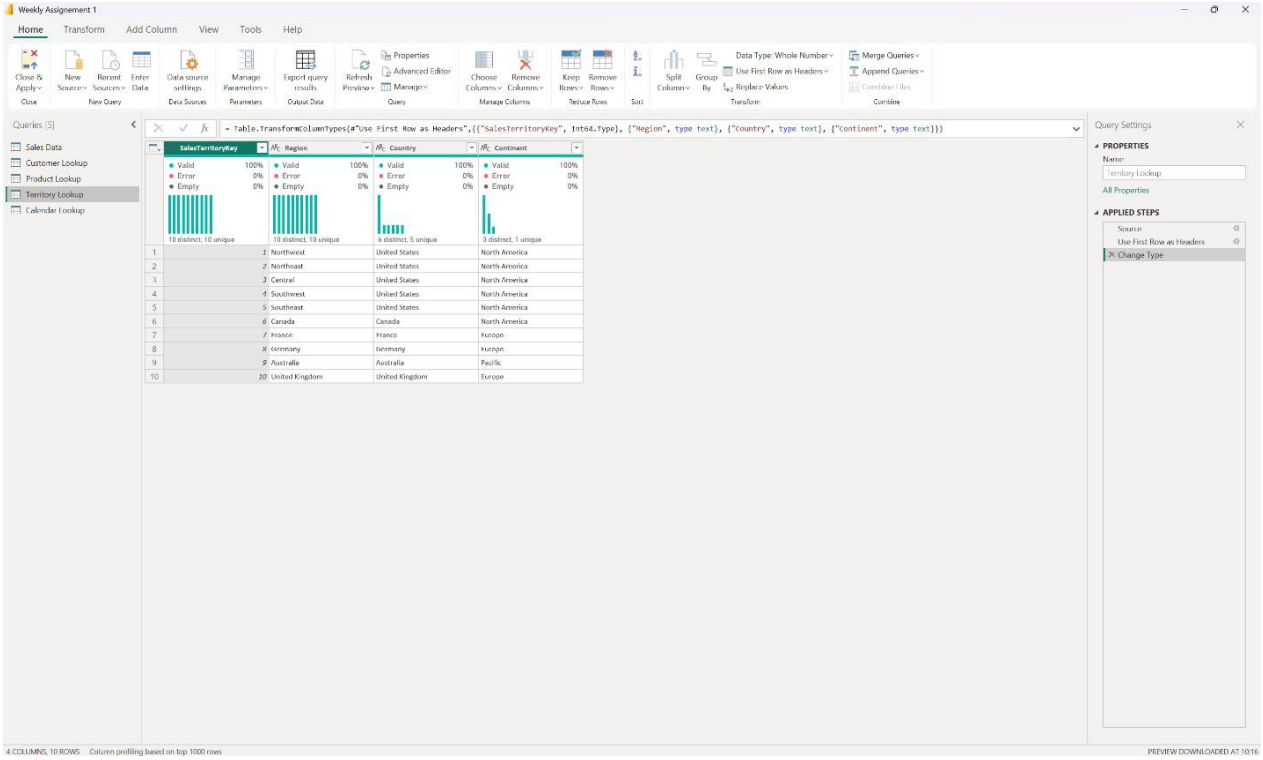


Figure 6: Territory Lookup Table

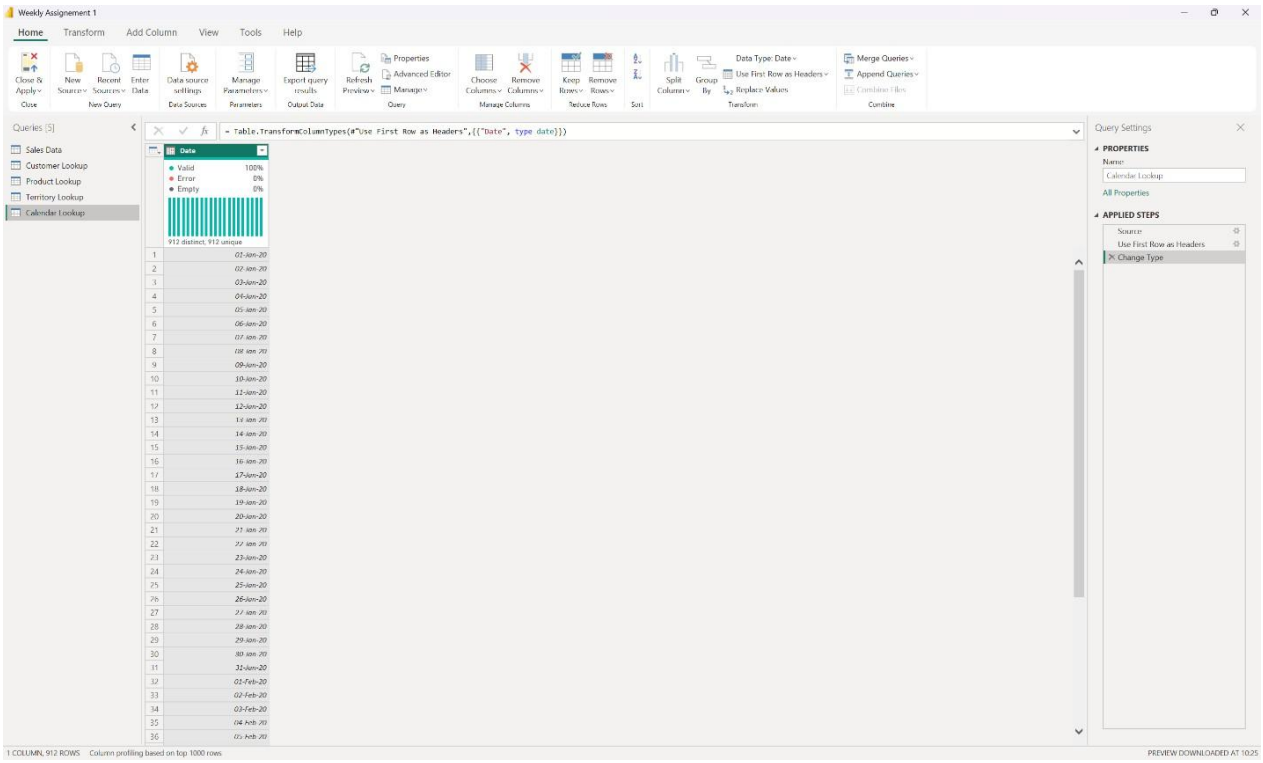


Figure 7: Calendar Lookup Table

Step 4: Data Modeling

After importing all tables, relationships were established between the fact table and lookup tables in the Model View to create a Star Schema:

- Sales → Customer Lookup (via CustomerKey)
- Sales → Product Lookup (via ProductKey)
- Sales → Territory Lookup (via TerritoryKey)
- Sales → Calendar Lookup (via OrderDate → Date)

This star schema ensures efficient filtering and accurate aggregation across all visuals in the dashboard.

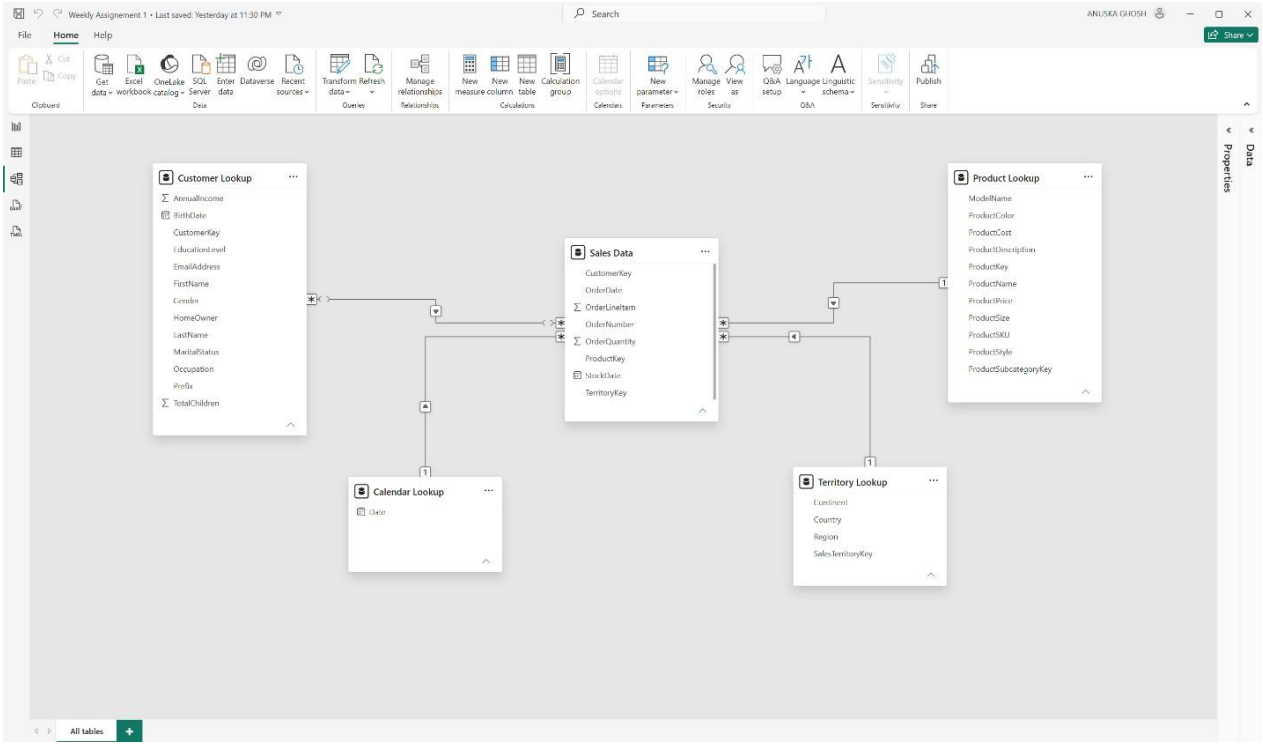


Figure 8: Data Model — Star Schema in Model View

Step 5: Dashboard Development

An interactive dashboard titled "Adventure Works Retail Sales Performance Dashboard" was developed in the Report View. The following visuals were created:

- **4 KPI Cards** - Total Orders (25.164K), Total Quantity Sold (84K), Total Customers (18.149K), Total Profit ($10.27M)
- **Sales by Product** - Clustered Bar Chart showing quantity sold per product
- **Sales by Customer Occupation** - Donut Chart showing sales distribution across customer segments
- **Sales Trend Over Time** - Line Chart showing yearly sales growth from 2020 to 2022
- **Sales by Store Location** - Filled Map showing geographical distribution of sales
- **2 Slicers** - Region and Year for interactive filtering

Output

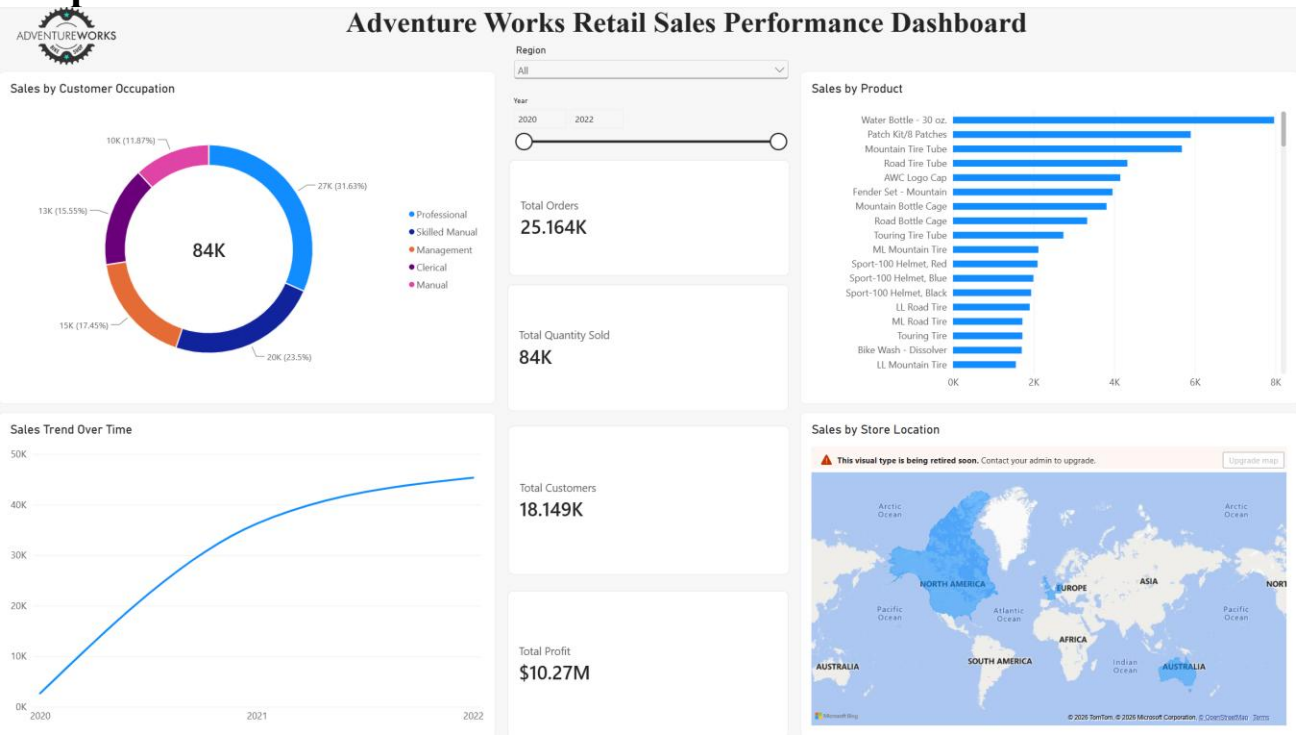


Figure 9: AdventureWorks Retail Sales Performance Dashboard — Report View

The final Power BI dashboard titled "AdventureWorks Sales Performance Dashboard" successfully visualizes the retail company's business data. The dashboard provides:

- **Total Orders:** 25.164K orders recorded across 2020-2022
- **Total Quantity Sold:** 84K units sold across all product categories
- **Total Customers:** 18.149K unique customers served
- **Total Profit:** $10.27M in net profit generated
- **Product Performance:** Water Bottle 30oz, Patch Kit, and Mountain Tire Tube are the top selling products
- **Customer Segments:** Professional customers contribute the highest sales at 31.63%
- **Sales Growth:** Consistent upward trend in sales from 2020 to 2022
- **Geographic Reach:** Sales recorded across North America, Europe, and Australia
- **Interactive Filters:** Region and Year slicers enable dynamic data exploration

Conclusion

This experiment successfully demonstrated the integration of data from multiple sources - CSV flat files and a MySQL database, into Power BI Desktop. The AdventureWorks dataset containing over 56,000 sales records was imported, cleaned, and modeled to create an interactive business dashboard. The star schema data model established efficient relationships between the fact and dimension tables, enabling accurate and meaningful visual analysis. The developed dashboard provides the retail company's management with a centralized view of sales performance across products, customers, regions, and time periods, supporting data-driven decision making.

Business Insights

The following key business insights were derived from the AdventureWorks Sales Performance Dashboard:

- **Top Selling Products:** Water Bottle 30oz, Patch Kit/8 Patches, and Mountain Tire Tube are the highest selling products by quantity, indicating strong demand for accessories and maintenance products.
- **Customer Segmentation:** Professional customers account for 31.63% of total sales, making them the most valuable customer segment for the company.
- **Profit Performance:** The business demonstrates a healthy profit margin, with total profit calculated based on the difference between product price and product cost across all sales transactions.
- **Sales Growth Trend:** Sales show a consistent upward trend from 2020 to 2022, with the steepest growth observed between 2021 and 2022, indicating business expansion.
- **Geographic Performance:** North America is the strongest performing region, followed by Europe and Australia, suggesting opportunities to expand in underperforming markets.
- **Order Volume:** With over 25,000 unique orders and 84,000 units sold, the company demonstrates strong transactional activity across the three-year period.
- **Interactive Analysis:** The Region and Year slicers enable management to drill down into specific territories and time periods for targeted business decisions.